

Writing Large Numbers in Scientific Notation*Question Bank 3.08*

Show your work. Circle your final answer.

For questions 1 through 4, write each number in scientific notation.

1. 9,000

2. 4,500,000

3. 310,000,000

4. 7,020,000,000

For questions 5 and 6, write each number in standard form.

5. 6.3×10^4

6. 1.05×10^8

Real quantities.

7. Write each quantity in scientific notation.

- a. There are about 1,670,000,000,000,000,000 molecules in a drop of water.
- b. The average distance from Earth to the Sun is about 93,000,000 miles.
- c. A computer displays the number of stars in a galaxy as **4E11**. Write it in standard form.

8. Complete each comparison.

- a. How many times larger is 8.1 than 5.4?
- b. How many times larger is 10^8 than 10^7 ?
- c. How many times larger is 8.1×10^8 than 5.4×10^7 ?

9. Answer each question.

a. A blue whale weighs about 4×10^5 pounds. An elephant weighs about 1×10^4 pounds. How many times heavier is the blue whale?

b. Florida has an area of about 6.5×10^4 square miles. Lake Okeechobee has an area of about 7.3×10^2 square miles. About how many times larger is Florida? Round to the nearest whole number.

10. Which of the following is greater? Explain how you know.

a. 9.5×10^5 or 1.2×10^6

b. 3.8×10^7 or 3.09×10^7

11. Which is 52,000,000 written in scientific notation?

- ☐ A. 5.2×10^7
- ☐ B. 5.2×10^8
- ☐ C. 5.2×10^6
- ☐ D. 52×10^6

12. Select ALL of the expressions that are equal in value to 3,400,000.

- ☐ A. 3.4×10^5
- ☐ B. 0.34×10^7
- ☐ C. 34×10^5
- ☐ D. 3.4×10^6
- ☐ E. 340×10^4
- ☐ F. 3.4×10^7

13. In question 12, only one of the four equal expressions is written in scientific notation. Which one is it, and why are the other three not?